Agent-Chronos: Hardware-Clock Speculative Transport Pacing and Causal Fencing for Cross-Cloud Multi-Agent Swarms

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***Abstract*—As enterprise deployments of autonomous agent swarms transition from single-tenant sandbox prototypes to heterogeneous multi-cloud topologies spanning Google Cloud Vertex AI, SAP S/4HANA, Salesforce Agentforce, and AWS Bedrock over Agent2Agent (A2A) and the Model Context Protocol (MCP), distributed consistency across external tool invocations degrades severely. Because Large Language Model (LLM) reasoning durations vary stochastically (200 ms to $5{,}000$ ms) and wide-area network (WAN) transit jitter (± 10 ms to 60 ms) reorders packets in flight, multi-agent systems exhibit high rates of causal inversion (19.50%) and phantom tool mutations (14.80%) when upstream speculative reasoning branches abort after downstream mutating actions have already been dispatched. Traditional distributed database locks (Two-Phase Commit, XA) cannot be held across third-party REST and gRPC endpoints, while post-hoc compensating sagas fail when external side-effects are irreversible. We present Agent-Chronos, a hardware-anchored L7 transport fencing and speculative pacing protocol that guarantees strict client-side egress linearizability and end-to-end causal delivery across Chronos-enabled multi-agent gateways without distributed database locks. Agent-Chronos: (1) embeds physical TrueTime atomic/GPS clock uncertainty tuples $\langle t_{\text{earliest}}, t_{\text{latest}}, \epsilon_u \rangle$, swarm-wide causal vectors $\mathcal{V}$, and per-destination sequence coordinates Seq_{dest} directly into HTTP/2 framing; (2) implements a two-phase user-space L7 egress pacer (prior to HPACK compression and TLS encryption) that stages speculative mutations during upstream verification (Δ_{stage}) and enforces a post-settlement TrueTime commit-wait barrier ($\Delta_{\text{wait}} = \max(0, t_{F,\text{latest}} + \Delta_{\text{abort_prop}} - t_{\text{now,earliest}})$) before wire release; (3) enforces an ingress causal hold barrier at receiving sidecars to neutralize WAN transit jitter while permitting dependency-free read-only queries to bypass pacing with 0.00 ms delay; and (4) provides pre-wire speculative revocation that purges aborted frames in user-space gateway queues so zero mutating payload bytes reach downstream endpoints. Across $N = 1{,}000$ multi-agent transactions per seed (5 seeds, $5{,}000$ total transaction runs per baseline) in an emulated cross-cloud testbed, Agent-Chronos reduced phantom mutations and causal inversions to 0.00%, bounding post-settlement TrueTime commit-wait pacing overhead to a mean of 5.91 ± 2.32 ms ($p_{50} = 5.90$ ms, $p_{95} = 9.80$ ms, $p_{99} = 10.84$ ms) while achieving a net end-to-end speculative speedup of 423.3 ms (34.6%) per committed transaction (388.5 ms, or 31.8%, unconditionally across all transactions including aborted branch retries) over non-speculative sequential execution. All protocol transitions are verifiable via deterministic HTTP/2 headers and audit trailers, providing end-to-end cross-organization protocol observability.**

***Index Terms*—Multi-Agent Systems, TrueTime, Causal Consistency, Transport Pacing, Linearizability, Speculative Execution, Model Context Protocol, Distributed Systems.**

I. Introduction

Autonomous multi-agent systems have emerged as a primary architectural paradigm for enterprise generative AI orchestration. Modern enterprise workflows coordinate swarms of specialized agents executing complex multi-turn objectives—such as reconciling financial ledgers, assessing procurement risk, and updating Enterprise Resource Planning (ERP) records across heterogeneous cloud boundaries [2], [3], [15].

However, as agent swarms expand across organizational and cloud boundaries, a critical distributed systems failure mode arises: **Speculative Causal Inversion and Phantom Tool Mutations**. To hide multi-second LLM generation latencies, modern agentic serving runtimes speculatively synthesize and stage downstream tool calls in parallel with upstream policy verification and sibling reasoning branches [10], [13], [14], [16]–[18]. Unlike deterministic microservices whose execution latency is tightly bounded, LLM reasoning and guardrail verification turns vary stochastically from 200 ms to over $5{,}000$ ms. Concurrently, regional WAN jitter across public interconnects (± 10 ms to 60 ms) reorders packets in flight. When a downstream agent dispatches an irreversible external mutation (e.g., executing an electronic funds transfer or releasing an ERP inventory hold) before an upstream speculative epoch settles—or when clock skew ($\pm \epsilon_u$) across distributed verification nodes causes an upstream abort signal and a downstream commit candidate to cross in flight—any subsequent upstream policy abort leaves a *phantom mutation* on the destination system of record.

Existing distributed transaction and coordination primitives do not fully resolve this boundary gap:

1) **Classical Distributed Locks, Agentic Sagas, and Logical Pre-Commit Staging:** Two-Phase Commit (2PC) and XA locks cannot be held across stateless third-party SaaS REST/gRPC APIs (e.g., Stripe, SAP NetWeaver, or Salesforce). Classical compensating sagas [4], multi-agent workflow validation coordinators such as SagaLLM [14] and ALAS [19], and post-facto undo runtimes such as GoEX [13] operate at the application orchestration layer without hardware-clock transport fencing. Similarly, recent speculative and transactional agent runtimes—including Speculative Actions [16], Sherlock [17], and Atomix [18]—overlap verification with logical pre-commit staging and read-

only bypass inside a single centralized orchestrator. However, when such logical pre-commit staging designs are deployed across *decentralized cross-node verification topologies* (modeled by our *Chronos Vector-Only Ablation* in Section IV), logical staging alone—without a hardware-anchored $2\epsilon_{\text{eff}}$ commit-wait barrier—leaks phantom mutations (2.90% in Table I) when distributed cross-node verification aborts race within the physical clock-skew window and incurs destination causal inversions (3.40%) across concurrent sibling branches ($\mathcal{V}_1 \parallel \mathcal{V}_2$).

2) **OS Speculative Buffering and NIC Pacers:** Operating-system speculative execution frameworks such as Speculator [10] buffer uncommitted kernel outputs inside a single OS domain, while end-host traffic shapers such as SENIC [12] and Carousel [11] pace NIC packet departure rates for bandwidth shaping. However, raw kernel TCP socket buffering (`sk_buff`) cannot selectively excise individual aborted tool frames multiplexed over a shared HTTP/2-over-TLS (`h2`) connection without corrupting stateful HPACK header tables, TLS 1.3 AEAD record sequence numbers, and TCP byte-stream sequence offsets.
3) **Logical and Hybrid Clocks Under Commodity NTP Skew:** Classical Lamport timestamps [5], Fidge/Mattern vector clocks [6], [7], and $\mathcal{O}(1)$-space Hybrid Logical Clocks (HLCs) [8] track message-passing causality, but under commodity NTP synchronization ($\epsilon_{\text{NTP}} \approx$ 25–100 ms) they incur wide physical uncertainty windows when enforcing real-time commit-wait barriers across independent gateways, while scalar HLCs cannot distinguish concurrent sibling branch forks without explicit per-agent vector coordinates.

To address these challenges, we introduce **Agent-Chronos**, an L7 transport fencing protocol that bridges hardware-anchored TrueTime uncertainty intervals [1] with modern agentic protocols (MCP [2] via FastMCP gateways and A2A [3]). Rather than claiming database-internal strict serializability over unmodified third-party storage engines that lack two-phase locking or Paxos consensus, Agent-Chronos formalizes and enforces **strict client-side egress linearizability** [9] (guaranteeing that no speculative mutation departs the egress gateway until its upstream epoch has deterministically settled and its physical TrueTime uncertainty and abort-propagation window has elapsed) together with **end-to-end causal delivery** across Chronos-enabled ingress sidecars (e.g., fronting enterprise SAP S/4HANA gateways). When invoking unmodified external third-party SaaS endpoints (such as Stripe) that cannot host an ingress sidecar, the sender-side gateway suppresses tombstone sequence ACKs, coordinates concurrent sibling writes via a shared logical egress gateway, and falls back to sequential per-resource in-flight pipelining ($+42.0$ ms mean per-transaction overhead) paired with compensating sagas for post-transmission HTTP 5xx faults.

II. System Architecture and Formal Invariants

Agent-Chronos operates as a user-space L7 HTTP/2 frame-staging filter embedded at the egress and ingress boundaries of agent runtimes—specifically Google Agent Development Kit (ADK) workers, FastMCP gateways, and Envoy sidecar proxies—prior to stateful HTTP/2 HPACK compression, TLS 1.3 record encryption, and kernel TCP `sendmsg()` socket calls.

A. TrueTime Uncertainty Representation and Multi-Cloud Clock Model

Let $\mathbb{N}_0 = \{0, 1, 2, \dots\}$ denote the non-negative integers and $\mathbb{N}^+ = \{1, 2, \dots\}$ denote the strictly positive integers. Let $t_{\text{real}} \in \mathbb{N}^+$ denote absolute physical coordinate time expressed in discrete integer nanoseconds since the Unix epoch. On any node u, a hardware-anchored TrueTime provider [1] backed by GPS receivers and rubidium atomic clocks exposes a closed physical time uncertainty interval around node u's local reading $t_{\text{now},u} = t_{\text{real}} + \delta_u$ (where $\delta_u \in [-\epsilon_u, +\epsilon_u]$ is node u's instantaneous signed physical clock offset):

$$T_{\text{now},u} = [t_{\text{now},u,\text{earliest}},\ t_{\text{now},u,\text{latest}}], \quad t_{\text{now},u,\text{latest}} - t_{\text{now},u,\text{earliest}} = 2\epsilon_u \tag{1}$$

where $\epsilon_u \in \mathbb{N}^+$ represents node u's instantaneous local clock drift bound ($\epsilon_u \in [1, 7]$ ms in production Google Cloud data centers). For wire serialization of a single node's raw clock sample $T_{\text{now},u}$, Agent-Chronos encodes the 3-tuple $\mathcal{W}(T_{\text{now},u}) = \langle t_{\text{now},u,\text{earliest}}, t_{\text{now},u,\text{latest}}, \epsilon_u \rangle$, where receivers verify the exact single-node framing identity $t_{\text{now},u,\text{latest}} - t_{\text{now},u,\text{earliest}} = 2\epsilon_u$. In particular, when an upstream verification node settles epoch E, its intra-cluster settlement token transmits $\mathcal{W}(T_E) = \langle t_{E,\text{earliest}}, t_{E,\text{latest}}, \epsilon_{\text{verifier}} \rangle$ to the egress gateway, which uses $\epsilon_{\text{verifier}}$ and its local $\epsilon_{\text{gateway}}$ to compose the heterogeneous cross-node uncertainty bound $\epsilon_{\text{eff}} = \lceil (\epsilon_{\text{verifier}} + \epsilon_{\text{gateway}})/2 \rceil \in \mathbb{N}^+$ (ceiling-rounded to preserve conservative integer nanosecond header encoding when $\epsilon_{\text{verifier}}$ and $\epsilon_{\text{gateway}}$ have opposite parity, while the Phase 2 barrier in Eq. (6) evaluates the exact integer sum $2\epsilon_{\text{eff}}^{\text{exact}} = \epsilon_{\text{verifier}} + \epsilon_{\text{gateway}} \in \mathbb{N}^+$; for the composite outbound settlement header `X-Agent-TrueTime-Window` $\langle t_{E,\text{earliest}}, t_{F,\text{latest}}, \epsilon_{\text{eff}} \rangle$, where $t_{F,\text{latest}}$ may be advanced beyond $t_{E,\text{latest}}$ by causal or per-key settlement floors in Eq. (5), downstream receivers verify the generalized framing invariant $t_{F,\text{latest}} - t_{E,\text{earliest}} > 0$, or $t_{F,\text{latest}} - t_{E,\text{earliest}} \geq 2\epsilon_{\text{verifier}}$ when $\epsilon_{\text{verifier}}$ is included in optional extended audit metadata). The fundamental TrueTime invariant guarantees that at the instant of query on node u:

$$t_{\text{now},u,\text{earliest}} \leq t_{\text{real}} \leq t_{\text{now},u,\text{latest}} \tag{2}$$

Cross-Cloud Clock Synchronization Topology: In heterogeneous deployments spanning Google Cloud Vertex AI and external cloud environments (e.g., AWS Bedrock, Salesforce Agentforce, or SAP S/4HANA), Agent-Chronos supports two deployment modes: (i) *Federated Egress Gateway Mode*, where external agent runtimes route both upstream epoch settlement signals and outbound MCP/A2A tool calls through a regional Google Cloud Apigee / Envoy Chronos gateway so that epoch commit timestamps T_E and egress commit-wait checks $T_{\text{now,gateway}}$ are anchored by the regional TrueTime cluster ($\epsilon_u \in [1, 7]$ ms). To prevent differential WAN transit jitter on the pre-gateway path from inverting external commit/abort signals prior to gateway arrival, either (a) external verifiers attach origin-signed PTP/NTP bounds $\mathcal{W}(T_{\text{origin}})$, in which case $\epsilon_{\text{eff}} = \lceil (\epsilon_{\text{origin}} + \epsilon_{\text{gateway}})/2 \rceil$ inherits the origin clock's uncertainty bound and—whenever abort-capable external verifiers communicate with the federated

gateway across a WAN without a unanimous pre-minting quorum—the effective Phase 2 abort-propagation margin in Eq. (6) incorporates the pre-gateway one-way WAN delay upper bound ($\Delta^{\text{eff}}_{\text{abort_prop}} = \Delta_{\text{abort_prop}} + \Delta_{\text{wan_pre}}$); or (b) the federated gateway requires a synchronous quorum ACK from participating verifiers of epoch E before minting T_E at its local TrueTime clock. Under option (b), when all abort-capable verifiers participate in a unanimous monotonic quorum (where no verifier can emit an abort after a positive ACK), unanimous ACK arrival alone precludes any racing abort with $\Delta_{\text{wan_pre}} = 0$ ms and unconditionally preserves the zero-byte wire invariant (this unanimous mode is mandatory for irreversible external endpoints); when a non-unanimous subset quorum or an asynchronous out-of-band guardrail can emit aborts independently of the quorum ACK channel, the gateway waits a conservative pre-gateway jitter margin $\Delta_{\text{wan_pre}}$ before minting T_E and assigns a monotonic quorum-epoch fence token, guaranteeing strict zero-byte pre-wire revocation as an operational scope condition under bounded pre-gateway network delay ($\leq \Delta_{\text{wan_pre}} + \Delta_{\text{wait}}$) while invoking deterministic post-release compensation as a fail-safe on reversible endpoints if a catastrophic WAN stall exceeds $\Delta_{\text{wan_pre}} + \Delta_{\text{wait}}$; and (ii) *Hardware PTP Sidecar Mode*, where non-Google cloud participants query local GPS/atomic Precision Time Protocol hardware clocks (e.g., AWS Time Sync Service `/dev/ptp0` with hardware-reported error bound $\epsilon_{\text{PTP},u} \in [1, 10]$ ms) at both verification and egress nodes and advertise $\epsilon_{\text{PTP},u}$ in $\mathcal{W}(T_{\text{now},u})$ (noting that when invoking unmodified third-party SaaS endpoints lacking an ingress sidecar across concurrent writers, distributed PTP sidecars still route through a single logical egress serialization gateway for that SaaS target).

B. Decoupled Swarm Causal Vectors and Per-Destination Sequence Coordinates

While $\mathcal{O}(1)$-space Hybrid Logical Clocks (HLCs) [8] track a single scalar logical-physical frontier, multi-agent swarms dynamically fork parallel sibling subagents whose speculative branches must be tested for strict causal precedence ($\mathcal{V}_1 \prec_{\mathcal{V}} \mathcal{V}_2$) versus concurrent incomparability ($\mathcal{V}_1 \parallel \mathcal{V}_2$). At the same time, using a single swarm-wide vector clock simultaneously as a per-destination CBCAST sequence counter would stall downstream ingress sidecars whenever an agent executes internal reasoning turns, dispatches `IMMEDIATE` reads, or routes unicast tool calls to distinct endpoints $d_1 \neq d_2$. Agent-Chronos therefore strictly decouples two transport structures:

- **Swarm-Wide Composite Causal Vector ($\mathcal{V}_i = \langle \mathbf{C}_i, t_{\mathbf{max},i} \rangle$):** Maintained by each agent $a_i \in$ AgentID, where $\mathbf{C}_i$: AgentID $\to \mathbb{N}_0$ is a Fidge/Mattern vector clock [6], [7] over active swarm agent IDs (with $\text{dom}(\mathbf{C}_i)$ denoting the finite non-zero support and missing keys defaulting to 0) and $t_{\text{max},i} \in \mathbb{N}_0$ (initialized to 0 and taking values in $\mathbb{N}^+$ after the first clock sample; equivalently written $\mathcal{V}_i.t_{\text{max}} \equiv t_{\text{max},i}$) is the maximum TrueTime upper bound (t_{latest}) observed across a_i's causal history. Evaluated strictly over the logical vector component $\mathbf{C}$ (with $t_{\text{max},i}$ serving as a commit-wait floor and physical tie-breaker for concurrent writes), strict causal precedence over $D = \text{dom}(\mathbf{C}_1) \cup \text{dom}(\mathbf{C}_2)$ is defined as $\mathcal{V}_1 \prec_{\mathcal{V}} \mathcal{V}_2 \iff (\forall k \in D,\, \mathbf{C}_1[k] \leq \mathbf{C}_2[k]) \wedge (\exists k \in D,\, \mathbf{C}_1[k] < \mathbf{C}_2[k])$, and concurrency as $\mathcal{V}_1 \parallel \mathcal{V}_2 \iff \neg(\mathcal{V}_1 \prec_{\mathcal{V}} \mathcal{V}_2) \wedge \neg(\mathcal{V}_2 \prec_{\mathcal{V}} \mathcal{V}_1)$.
- **Per-Destination Transport Sequence Coordinate ($\mathbf{Seq_{dest}}[d, a_i]$):** For each destination endpoint d, the egress gateway maintains a dedicated gap-free monotonic counter $\text{Seq}_{\text{dest}}[d, a_i] \in \mathbb{N}_0$ (initialized to 0 and incremented to $1, 2, \cdots \in \mathbb{N}^+$ upon Phase 2 release) and a compact destination dependency map $\text{Dep}_{d,i}$: AgentID $\to \mathbb{N}_0$ (with missing keys defaulting to 0) recording the highest destination-d sequence numbers causally preceding the current frame.

1) **Two-Stage Pre-Staging vs. Post-Phase-2 Sequence Assignment:** Parallel speculative branches ($\mathcal{V}_1 \parallel \mathcal{V}_2$) fork across distinct logical subagent identifiers $a_1 \neq a_2 \in$ AgentID (receiving independent vector entries $\mathbf{C}[a_1], \mathbf{C}[a_2]$). When a single agent $a_i \in$ AgentID initiates a mutating tool frame F (`F.is_mutation` = true) or an outbound A2A delegation turn at local TrueTime interval $T_{\text{now},i} = [t_{\text{now},i,\text{earliest}}, t_{\text{now},i,\text{latest}}]$, it constructs an isolated branch-local tentative pre-staging vector $\mathcal{V}_F = \langle \mathbf{C}_F, t_{\text{max},F} \rangle$ from its committed state $\langle \mathbf{C}_i, t_{\text{max},i} \rangle$ using an atomic per-agent tentative ticket counter $c_{\text{tent},i} \leftarrow c_{\text{tent},i} + 1$ (whereas exploratory read-only queries with `F.is_mutation` = false inspect $\langle \mathbf{C}_i, t_{\text{max},i} \rangle$ without incrementing $c_{\text{tent},i}$, and overlapping same-agent mutations to the same destination/key are chained in ticket order $c_{\text{tent},1} < c_{\text{tent},2}$ at Phase 2 release; if F is revoked in Rule 4, $\mathcal{V}_F$ is simply discarded without mutating a_i's shared committed state $\langle \mathbf{C}_i, t_{\text{max},i} \rangle$, and any numerical gap in committed $\mathbf{C}_i[a_i]$ after a later ticket commits is benign because vector precedence $\prec_{\mathcal{V}}$ tests $\leq$ / $<$ while gap-free ingress FIFO delivery relies exclusively on $\text{Seq}_{\text{dest}}[d, a_i]$):

$$\begin{aligned} \mathbf{C}_F &\leftarrow \mathbf{C}_i[a_i \mapsto c_{\text{tent},i}], \\ t_{\text{max},F} &\leftarrow \max(t_{\text{max},i},\, t_{\text{now},i,\text{latest}}). \end{aligned} \tag{3}$$

Crucially, the per-destination sequence counter $\text{Seq}_{\text{dest},F}[d, a_i] \leftarrow \text{Seq}_{\text{dest}}[d, a_i] + 1$ is assigned and incremented **strictly upon Phase 2 commit-wait release** of a mutating frame addressed to d, at which point a_i's shared committed causal frontier atomically joins $\mathbf{C}_i \leftarrow \mathbf{C}_i \sqcup \mathbf{C}_F$ and $t_{\text{max},i} \leftarrow \max(t_{\text{max},i}, t_{F,\text{latest}})$ and records $\text{Seq}_{\text{dest},F}[d, a_i]$ in $\text{Dep}_{d,i}[a_i]$; meanwhile, the outbound frame F itself serializes its pre-staged dependency snapshot $\mathcal{V}_F = \langle \mathbf{C}_F, t_{F,\text{latest}} \rangle$ in `X-Agent-Causal-Fence-Vector` and its pre-staged $\text{Dep}_{d,F}$ (updated only with prior committed same-agent releases to d and explicitly chained Rule 1/sibling writes) in `X-Agent-Dest-Seq`, preventing unrelated inbound responses arriving at a_i during F's Phase 1/Phase 2 hold from retroactively injecting false causal dependencies into F. Because $t_{\text{now},u} = t_{\text{real}} + \delta_u$ with $\delta_u \in [-\epsilon_u, +\epsilon_u]$ gives $t_{\text{now},u,\text{latest}} = t_{\text{real}} + \delta_u + \epsilon_u \leq t_{\text{real}} + 2\epsilon_u$ (attained when node u's clock runs fast at maximum drift $\delta_u = +\epsilon_u$), and because Phase 2 waits until $t_{\text{real}} \geq t_{F,\text{latest}}$ while unpaced local/A2A turns in Eqs. (3)–(4) take the pointwise maximum of existing $t_{\text{now},u,\text{latest}}$ bounds rather than summing ϵ_u across hops, the physical lead of $t_{\text{max},i}$ is strictly non-accumulating across deep multi-agent

delegation chains—satisfying $t_{\max,i} - t_{\text{real}} \leq \max_u \epsilon_u$ at $\mathbb{E}[\delta_u] = 0$ and the deterministic worst-case bound $t_{\max,i} - t_{\text{real}} \leq \max(2\epsilon_{\text{src}} - d_{\min}, 2\epsilon_i) \leq 2\max_u \epsilon_u$ (for minimum inter-node transit delay $d_{\min} > 0$) without inflating downstream Δ_{floor}. To ensure transitive causal ordering across A2A delegations, any outbound A2A task delegation from a_i to subagent a_j that depends on an uncommitted write F_1 in epoch E either awaits Phase 2 release of F_1 so $\text{Dep}_{d,i}$ carries the committed $\text{Seq}_{\text{dest},F_1}[d, a_i]$, or propagates a tentative epoch dependency handle (E, F_1) so a_j's downstream calls inherit E's Phase 1/Phase 2 barrier; upon F_1's Phase 2 release, the releasing gateway broadcasts F_1's newly assigned $\text{Seq}_{\text{dest},F_1}[d, a_i]$ and committed $\mathcal{V}_{F_1}$ to a_j's egress gateway to merge into $\text{Dep}_{d,F_2}[a_i]$ and $\mathcal{V}_{F_2}$ before F_2 releases from Phase 2. Because frames revoked pre-wire during Phase 1 or Phase 2 are purged before $\text{Seq}_{\text{dest}}[d, a_i]$ is incremented, pre-wire revocations never create sequence gaps and never transmit traffic toward downstream targets; payload-free tombstone sequence ACKs are emitted only if a post-release transport stream resets mid-flight on a sidecar-fronted link (and are suppressed for unmodified SaaS endpoints).

2) **Causal Merge Rule:** When agent a_j receives an A2A or tool response frame carrying $\mathcal{V}_i = \langle \mathbf{C}_i, t_{\max,i} \rangle$ and per-destination watermarks $\text{Dep}_{d,i}$, it merges $\mathcal{V}_i$ via the pointwise join operator ($\sqcup$) across $\text{dom}(\mathbf{C}_j) \cup \text{dom}(\mathbf{C}_i)$:

$$\mathcal{V}_j.\text{merge}(\mathcal{V}_i) = \langle \mathbf{C}_j \sqcup \mathbf{C}_i,\ \max(t_{\max,j},\ t_{\max,i}) \rangle\ , \quad (4)$$

where $(\mathbf{C}_j \sqcup \mathbf{C}_i)[k] \triangleq \max(\mathbf{C}_j[k], \mathbf{C}_i[k])$, and merges $\text{Dep}_{d,j}[k] \leftarrow \max(\text{Dep}_{d,j}[k], \text{Dep}_{d,i}[k])$. Mapping active swarm agent IDs ($|\text{AgentID}| \leq 16$) to a 16-element positional integer array in compact JSON {`"c":[c_0,...,c_15],"t":t_max,"b":bitmap`} (with per-session counters $c_k < 10^4$, a 19-digit nanosecond integer `t`, and a 16-bit integer bitmap `b` ≤ 65535 encoding active colliding sibling forks as defined in Section II-D) requires 69–121 bytes of unformatted raw JSON (117–121 bytes at 4-digit counters, or up to 142–184 bytes when including optional whitespace/epoch metadata, remaining ≤ 190 bytes) and ≤ 256 bytes ($4\lceil 190/3 \rceil$) when Base64-encoded in HTTP/2 headers.

C. Two-Phase L7 Speculative Staging and TrueTime Commit-Wait Pacing

Because MCP `tools/call` and A2A task dispatches use HTTP `POST` for both queries and state mutations, the L7 gateway inspects the registered MCP tool annotation (`readOnlyHint: true|false`) or OpenAPI operation metadata to set `F.is_mutation` $\in$ {true, false}, extracts the composite target resource key-set $\mathcal{K}(F) \subseteq \mathcal{K}$ via registered per-tool JSONPath selectors over MCP `params.arguments`, A2A task/artifact payloads, or OpenAPI path/query/body parameters (e.g., {`$.ledger_id`}, multi-key {`$.debit_id`, `$.credit_id`} for atomic transfers, or fallback universal wildcard $\{\star\}$ for unmapped or range/aggregate queries, where $\{\star\}$ intersects every non-empty key-set $\mathcal{K}(F')$ at destination d, $\{\star\} \cap \mathcal{K}(F') = \mathcal{K}(F') \neq \emptyset$, consults $\max_k t_{\text{key_floor}}[d, k]$ and destination-wide floor $t_{\text{key_floor}}[d, \star]$, and maps to a destination-wide root lock ordered canonically before specific resource keys), and reads the upstream speculative epoch identifier E and causal vector $\mathcal{V}_F$ from the request context. Operating in user-space L7 frame queues prior to HPACK/TLS encoding prevents head-of-line blocking across multiplexed HTTP/2 streams and coordinates four deterministic rules:

1) **Conditional Safe Read-Only Bypass:** If `F.is_mutation` = false and neither (a) a causally preceding write F_{prior} ($\mathcal{V}_{F_{\text{prior}}} \prec_{\mathcal{V}} \mathcal{V}_F$) across Phase 1 (`STAGE_HELD`) or Phase 2 (`PACED_HELD`) nor (b) a concurrent sibling write F_{sib} currently in Phase 2 (`PACED_HELD`) with an earlier or overlapping commit interval $t_{F_{\text{sib}},\text{earliest}} \equiv t_{E_{\text{sib}},\text{earliest}} \leq t_{\text{now},F,\text{latest}}$ (and, on unmodified external SaaS endpoints lacking an ingress sidecar, no overlapping in-flight write currently holding a sender-side per-resource SaaS pipeline lock on $\mathcal{K}(F)$) shares an overlapping resource key ($\mathcal{K}(F_{\text{prior/sib}}) \cap \mathcal{K}(F) \neq \emptyset$), pacing status is set to `IMMEDIATE` and the frame is dispatched with 0.00 ms pacing delay. To ensure unsettled speculative sibling writes in Phase 1 (`STAGE_HELD`) linearize strictly after concurrent exploratory `IMMEDIATE` reads without stalling reads for 200–5,000 ms, dispatching an `IMMEDIATE` read F advances the monotonic per-key floor $t_{\text{key_floor}}[d, k] \leftarrow \max(t_{\text{key_floor}}[d, k], t_{\text{now},F,\text{latest}} + 1\text{ ns})$ for all $k \in \mathcal{K}(F)$ (while Phase 2 visibility is evaluated over the egress gateway's locally observed settlement state, replicated across distributed PTP sidecars via intra-cluster settlement tokens). Otherwise, F enters `SAFE_READ_WAIT` and is held at the egress gateway until the blocking write (F_{prior} or F_{sib}) transitions to `RELEASED` and assigns its sequence coordinate $\text{Seq}_{\text{dest},F_{\text{prior/sib}}}[d, a_{\text{prior/sib}}]$, which is merged into $\text{Dep}_{d,F}$ prior to dispatching F. If a causally preceding write F_{prior} (case a) is revoked via Rule 4, the dependent read F is aborted with HTTP 409 Conflict; conversely, if a concurrent sibling write F_{sib} (case b, $\mathcal{V}_{F_{\text{sib}}} \parallel \mathcal{V}_F$) is revoked pre-wire during Phase 2, F_{sib} never takes effect and F is immediately unblocked and re-evaluated for `IMMEDIATE` dispatch. Furthermore, on sidecar-fronted links, the egress gateway merges the sequence coordinate $\text{Seq}_{\text{dest},F_{\text{sib}}}[d, a_{\text{sib}}]$ of any overlapping chained or recently released sibling write (within $\Delta_{\text{jitter_hold}}$) into $\text{Dep}_{d,F}$, and any read F with non-empty $\text{Dep}_{d,F}$ attaches $\langle \text{Seq}_{\text{dest},F} = 0, \text{Dep}_{d,F} \rangle$ in `X-Agent-Dest-Seq` so the ingress sidecar holds F until $\forall a_k \in \text{dom}(\text{Dep}_{d,F})$, $\text{Seq}_{\text{loc},d}[a_k] \geq \text{Dep}_{d,F}[a_k]$, preventing WAN jitter from reordering a post-release read ahead of an in-flight released write.

2) **Phase 1 — Speculative Epoch Staging (Δ_{stage}):** If `F.is_mutation` = true and F depends on an active upstream speculative reasoning or policy-verification branch E (duration 200–5,000 ms), the L7 filter holds F in user-space staging memory (`STAGE_HELD`) while upstream generation and parallel policy verification execute concurrently. Once upstream epoch E com-

pletes verification and commits its decision at verifier TrueTime interval $T_E = [t_{E,\text{earliest}}, t_{E,\text{latest}}]$ (with verifier uncertainty $\epsilon_{\text{verifier}}$), the epoch broadcasts an intra-cluster settlement token $\mathcal{W}(T_E)$ containing $t^*_{E,\text{latest}} = \max(t_{E,\text{latest}}, t_{\max,F})$, and the gateway computes the frame's monotonic commit upper bound $t_{F,\text{latest}}$ by incorporating the per-key settlement floor $t_{\text{key_floor}}[d,k] \in \mathbb{N}_0$ across all $k \in \mathcal{K}(F)$ (alongside the destination-wide wildcard floor $t_{\text{key_floor}}[d,\star]$):

$$t_{F,\text{latest}} = \max\left(t^*_{E,\text{latest}},\, t_{\text{key_floor}}[d,\star],\, \max_{k\in\mathcal{K}(F)} t_{\text{key_floor}}[d,k]\right), \tag{5}$$

immediately advancing $t_{\text{key_floor}}[d,k] \leftarrow \max(t_{\text{key_floor}}[d,k], t_{F,\text{latest}} + 1\text{ ns})$ for all $k \in \mathcal{K}(F)$ upon Phase 1 settlement (and replicating the updated floor across distributed PTP sidecars in the settlement token so that the intra-cluster propagation margin $\Delta_{\text{abort_prop}}$ during Phase 2 absorbs any cross-sidecar replication lag before the sibling bitmap is finalized at Phase 2 release).

3) **Phase 2 — TrueTime Commit-Wait Serialization (Δ_{wait}):** Upon receiving the settlement token for epoch E, the L7 egress pacer transitions F to `PACED_HELD` and halts wire transmission until its local TrueTime lower bound $t_{\text{now,earliest}} = t_{\text{now,gateway}} - \epsilon_{\text{gateway}}$ strictly surpasses $t_{F,\text{latest}} + \Delta_{\text{abort_prop}}$, where $\Delta_{\text{abort_prop}}$ is the upper bound on intra-cluster control-plane abort propagation delay from any sibling verification node ($\Delta_{\text{abort_prop}} \le 0.5$ ms over intra-cluster gRPC, or 0 ms under co-located quorum settlement):

$$\begin{aligned}\Delta_{\text{wait}} &= \max(0,\, t_{F,\text{latest}} + \Delta_{\text{abort_prop}} - t_{\text{now,earliest}}),\\ \mathbb{E}[\Delta_{\text{wait}}] &\approx \max(0,\, 2\epsilon_{\text{eff}} + \Delta_{\text{abort_prop}} - \Delta t_{\text{ipc}})\end{aligned} \tag{6}$$

where the first line is the **normative runtime barrier** evaluated by polling local TrueTime hardware (since under signed local clock offsets $\delta_u \in [-\epsilon_u, +\epsilon_u]$, physical IPC transit $\Delta_{\text{real_ipc}} \triangleq t_{\text{real,now}} - t_{\text{real},E}$, and combined causal/key floor $\Delta_{\text{floor}} = t_{F,\text{latest}} - t_{E,\text{latest}} \ge 0$, the instantaneous wait equals $\max(0, \Delta_{\text{floor}} + 2\epsilon_{\text{eff}} + (\delta_{\text{verifier}} - \delta_{\text{gateway}}) + \Delta_{\text{abort_prop}} - \Delta_{\text{real_ipc}}))$, and the second line gives the tight Jensen's inequality approximation ($\mathbb{E}[\max(0,X)] \approx \max(0, \mathbb{E}[X])$, exact to $< 0.4\%$ zero-clamping) under zero-mean clock offsets ($\mathbb{E}[\delta_{\text{verifier}} - \delta_{\text{gateway}}] = 0$) and $\Delta_{\text{floor}} = 0$, where $\Delta t_{\text{ipc}} \triangleq \mathbb{E}[\Delta_{\text{real_ipc}}] = \mathbb{E}[t_{\text{now,gateway}} - t_E]$ denotes expected IPC transit relative to the raw settlement midpoint $t_E \triangleq t_{E,\text{latest}} - \epsilon_{\text{verifier}}$ (which coincides with the floor-adjusted midpoint $t^*_E \triangleq t_{F,\text{latest}} - \epsilon_{\text{verifier}}$ when $\Delta_{\text{floor}} = 0$). A conflicting causal abort emitted by a sibling verifier at TrueTime interval $T_{\text{abort}} = [t_{\text{abort,earliest}}, t_{\text{abort,latest}}]$ preempts epoch E under the conservative interval-overlap condition $t_{\text{abort,earliest}} \le t_{F,\text{latest}}$ (which captures every abort physically emitted at $t_{\text{real,abort}} \le t_{F,\text{latest}}$, since $t_{\text{abort,earliest}} \le t_{\text{real,abort}}$ by Eq. (2)): waiting until $t_{\text{now,earliest}} \ge t_{F,\text{latest}} + \Delta_{\text{abort_prop}}$ guarantees by Eq. (2) that $t_{\text{real}} \ge t_{F,\text{latest}} + \Delta_{\text{abort_prop}} \ge t_{\text{real,abort}} + \Delta_{\text{abort_prop}}$, ensuring that any such abort has both been emitted and physically arrived at the egress pacer before F is released.

4) **Pre-Wire In-Flight Revocation:** If epoch E aborts during Phase 1 (Δ_{stage}) or if a cross-node policy abort satisfying $t_{\text{abort,earliest}} \le t_{F,\text{latest}}$ arrives within the Phase 2 commit-wait window (Δ_{wait}), the unencoded frame F is immediately purged from the user-space L7 queue before any bytes enter the HPACK encoder, TLS record layer, or kernel TCP socket buffer. The gateway returns a deterministic HTTP 409 Conflict response to the calling agent, ensuring **zero mutating payload bytes** reach the downstream SaaS endpoint.

D. Ingress Causal Re-Sequencing Barrier Against WAN Jitter

While sender-side commit-wait pacing orders departure timestamps in t_{real}, public WAN transit jitter (± 10 ms to 60 ms) can reorder packets in flight. For endpoints fronted by an Agent-Chronos ingress sidecar or enterprise API gateway (e.g., fronting SAP S/4HANA OData/RFC connectors), the ingress filter maintains per-destination applied sequence watermarks $\text{Seq}_{\text{loc},d}$ and buffers any arriving mutating frame F from sender a_s until $\text{Seq}_{\text{loc},d}[a_s] = \text{Seq}_{\text{dest},F}[d,a_s] - 1$ and $\forall a_k \in \text{dom}(\text{Dep}_{d,F})$, $\text{Seq}_{\text{loc},d}[a_k] \ge \text{Dep}_{d,F}[a_k]$ (and similarly holds any dependent read carrying $\text{Dep}_{d,F}$ until $\forall a_k \in \text{dom}(\text{Dep}_{d,F})$, $\text{Seq}_{\text{loc},d}[a_k] \ge \text{Dep}_{d,F}[a_k]$). For concurrent sibling writes ($\mathcal{V}_1 \parallel \mathcal{V}_2$) where F_1 (from a_{s1}) precedes F_2 (from a_{s2}) in deterministic ($t_{F,\text{latest}}$, AgentID) order ($t_{F_1,\text{latest}} < t_{F_2,\text{latest}}$, or equality broken by $\text{AgentID}(a_{s1}) < \text{AgentID}(a_{s2})$), the monotonic per-key settlement floor $t_{\text{key_floor}}[d,k]$ in Eq. (5) guarantees that any sibling settling Phase 1 after F_1 receives $t_{F_2,\text{latest}} > t_{F_1,\text{latest}}$. Upon Phase 2 release of the later-ordered write F_2, the egress gateway finalizes the 16-bit sibling fork bitmap `b` in `X-Agent-Causal-Fence-Vector`, setting bits *strictly* for concurrent sibling agents a_{s1} that have already settled Phase 1 with an earlier ($t_{F_1,\text{latest}}$, AgentID) order and overlapping resource keys ($\mathcal{K}(F_1) \cap \mathcal{K}(F_2) \neq \emptyset$) targeting d (supplemented by F_1's sequence coordinate tag in `X-Agent-Dest-Seq` when a_{s1} has multiple pipelined writes, and emitting a lightweight control-plane `SIBLING_CLEAR` frame if a flagged F_1 is revoked during Phase 2, so aborted, later-settled, read-only, or disjoint-key sibling branches never set bits in F_2's bitmap). Upon ingress arrival of F_2 once $t_{\text{now,earliest}} \ge t_{F_2,\text{latest}}$, the ingress sidecar initiates a bounded WAN jitter countdown $\Delta_{\text{jitter_hold}}$ *if and only if* F_2's bitmap `b` contains at least one bit for an earlier-ordered sibling write F_1 that has neither been applied at the ingress sidecar nor cleared by `SIBLING_CLEAR`; this hold terminates **immediately** (0 ms further wait) as soon as all flagged earlier sibling writes F_1 arrive and are applied (or cleared), waiting the full $\Delta_{\text{jitter_hold}} = (t_{\text{wan,max}} - t_{\text{wan,min}}) + 2\epsilon_{\text{gateway_diff}} + \Delta_{\text{abort_prop}}$ (which bounds maximum differential one-way WAN transit plus cross-gateway clock/replication offset after F_1's latest Phase 2 release, reducing to $t_{\text{wan,max}} - t_{\text{wan,min}} = 60$ ms under a shared regional egress gateway) only when F_1 is pathologically delayed. Conflicting writes are applied in deterministic ($t_{F,\text{latest}}$, AgentID) order once $t_{\text{now,earliest}} \ge t_{F,\text{latest}}$. If a pathologically delayed earlier sibling write F_1 arrives after F_2's $\Delta_{\text{jitter_hold}}$ expires and F_2 has already been applied with $t_{F_1,\text{latest}} < t_{F_2,\text{latest}}$, the ingress sidecar rejects F_1 with deterministic HTTP 409 Stale Sibling Timestamp while advancing

$\text{Seq}_{\text{loc},d}[a_{s1}]$ to $s_{\text{tomb}} = \text{Seq}_{\text{dest},F_1}[d, a_{s1}]$ as a tombstoned coordinate so subsequent frames from a_{s1} never stall, and cascades HTTP 409 aborts only to unapplied in-flight frames carrying $\text{Dep}_{d,F}[a_{s1}] \geq s_{\text{tomb}}$ whose acknowledged-tombstone watermark in `X-Agent-Dest-Seq` satisfies $\text{AckTomb}_{d,F}[a_{s1}] < s_{\text{tomb}}$ (whereas post-409 recovery writes synthesized after the agent processes the rejection carry $\text{AckTomb}_{d,F}[a_{s1}] \geq s_{\text{tomb}}$ and execute normally). For unmodified external third-party SaaS APIs without an ingress sidecar, the sender-side gateway acquires per-resource pipeline locks in $\mathcal{K}(F)$ upon Phase 2 transition to `RELEASED` in deterministic $(t_{F,\text{latest}}, \text{AgentID})$ order (and across multi-key sets in canonical lexicographical key order to prevent deadlocks), holds any dependent write or read F_2 ($\mathcal{K}(F_1) \cap \mathcal{K}(F_2) \neq \emptyset$) until the HTTP 2xx acknowledgment of F_1 returns (or a bounded lock-lease timeout on gateway crash/timeout reconciles F_1's terminal idempotency-key status and completes compensating saga recovery before releasing the lock to F_2), and pairs post-transmission HTTP 5xx faults with compensating sagas [4].

III. Wire Protocol and Observability Invariants

Internal pre-wire gateway queue states (`STAGE_HELD`, `PACED_HELD`) resolve to four terminal wire dispositions ({`IMMEDIATE`, `SAFE_READ_WAIT`, `RELEASED`, `REVOKED`}) prior to wire emission (while sidecar-to-sidecar control notifications for Phase 2 sibling revocations and post-release stream resets use zero-payload HTTP/2 `HEADERS`-only control frames with `END_STREAM=1` and `X-Agent-Pacing-Status` set to `SIBLING_CLEAR` or `TOMBSTONE_ACK`, which are terminated and consumed strictly at the Chronos ingress sidecar without forwarding any request to the backend ERP/SaaS service). Each speculative unary or client-streaming gRPC/MCP mutation either opens a dedicated multiplexed HTTP/2 stream whose initial `HEADERS` and request `DATA` frames are held in user-space until Phase 2 completes, or—on long-lived bidirectional MCP/A2A streams—holds individual JSON-RPC request `DATA` frames in user-space `STAGE_HELD`/`PACED_HELD` queues prior to wire emission (carrying per-call `X-Agent-*` settlement fields inside the L7 JSON-RPC `_meta` envelope upon Phase 2 release, while session-closing HTTP/2 `END_STREAM` trailers emit a cumulative session audit digest), strictly preserving the zero-byte wire invariant upon revocation without prematurely half-closing active bidirectional streams. Agent-Chronos injects four core settlement/-pacing HTTP/2 headers (plus `X-Agent-Dest-Seq` carrying $\langle \text{Seq}_{\text{dest},F}[d, a_s], \text{Dep}_{d,F}, \text{AckTomb}_{d,F} \rangle$ on mutating calls and dependent reads, totaling five `X-Agent-*` headers):

- `X-Agent-TrueTime-Window`: Serialized settlement 3-tuple $\langle t_{E,\text{earliest}}, t_{F,\text{latest}}, \epsilon_{\text{eff}} \rangle$ in integer nanoseconds, where $t_{F,\text{latest}}$ is given by Eq. (5), $\epsilon_{\text{eff}} = \lceil (\epsilon_{\text{verifier}} + \epsilon_{\text{gateway}})/2 \rceil$, and $t_{F,\text{latest}} - t_{E,\text{earliest}} > 0$.
- `X-Agent-Causal-Fence-Vector`: Compact base64/JSON serialization ($\leq$ 256 B) of pre-staged $\mathcal{V}_F = \langle \mathbf{C}_F, t_{F,\text{latest}} \rangle$ and 16-bit active preceding sibling fork bitmap `b`.
- `X-Agent-Pacing-Status`: Terminal disposition in {`IMMEDIATE`, `SAFE_READ_WAIT`, `RELEASED`, `REVOKED`} (plus sidecar-internal control dispositions `SIBLING_CLEAR`, `TOMBSTONE_ACK`).
- `X-Agent-Pacing-Delay-Ms`: Post-settlement gateway hold duration in milliseconds (Δ_{wait} for `RELEASED` mutations, chained wait for `SAFE_READ_WAIT` reads, and elapsed Phase 2 hold prior to abort for `REVOKED` frames; Phase 1 Δ_{stage} is exported in OpenTelemetry attribute `chronos.stage_delay_ms`).

Dual-Plane Observability of Revoked Frames: When a frame is purged via Pre-Wire In-Flight Revocation, zero mutating bytes are transmitted to the external SaaS target. Instead, `X-Agent-Pacing-Status: REVOKED` and `X-Agent-Pacing-Delay-Ms` are emitted on (i) the synchronous HTTP 409 response wire back to the upstream orchestrator and (ii) a payload-free signed telemetry trailer exported to the cross-organization OpenTelemetry audit sink, ensuring 100% protocol auditability without leaking aborted payloads to third-party APIs.

IV. Empirical Evaluation

A. Experimental Testbed and Workload Methodology

We evaluated Agent-Chronos on an emulated cross-cloud HTTP/2 Envoy + FastMCP testbed modeling multi-agent financial ledger and procurement workflows (adapted from τ-bench [15] tool-invocation schemas) between Google Cloud Vertex AI and an SAP S/4HANA ERP gateway across $N = 1{,}000$ multi-agent transactions per seed (4,200 total tool frames per run: 2,520 [60%] mutating frames and 1,680 [40%] read-only queries over a $|\mathcal{K}| = 64$ resource key-space under Zipf($s = 1.15$) contention, evaluated across 5 independent random seeds [5,000 total transactions per baseline] with a controlled fault-injection budget of 148 aborted transactions per seed [14.80%, 0.0 across-seed SD by design to isolate protocol behavior under a fixed fault rate] and stochastic network/clock draws across seeds). The 6-agent hierarchical DAG topology comprises 1 root orchestrator, 2 tier-1 domain planners, and 3 tier-2 execution subagents (depth 3, max branching factor 2, baseline $t_{\text{exec}} = 650.0 \pm 161.2$ ms mean tool/agent execution path, which decomposes into $t_{\text{prep}} = 560.0 \pm 159.3$ ms of upstream agent reasoning, tool synthesis, and staging plus $t_{\text{rtt}} = 90.0 \pm 24.5$ ms of post-release final commit WAN RTT), where each transaction dispatches 1 to 2 exploratory read queries and 2 to 3 mutating tool calls (mean 4.20 frames/transaction, with 46.6% of transactions containing dependent same-key multi-write chains and 5.8% [97.4/1,680] of reads issuing post-settlement verification queries during an active Phase 2 `PACED_HELD` window on the same key), with speculative verification guarding the terminal mutating commit frame of each transaction. Network links were configured with Linux `tc netem` around a 90 ms base RTT (45 ms one-way base delay with $\pm$30 ms independent uniform per-direction jitter, yielding Triangular(-60 ms, 0 ms, $+60$ ms) RTT jitter, strictly positive one-way transit $\in [15, 75]$ ms, and ingress sibling hold bound $\Delta_{\text{jitter_hold}} = 60$ ms matching the $75 - 15 = 60$ ms max one-way spread; co-located regional verifiers use $\Delta_{\text{wan_pre}} = 0$ ms), independent verifier and egress gateway TrueTime drift bounds $\epsilon_{\text{verifier}}, \epsilon_{\text{gateway}}$ i.i.d. $\sim$ Uniform(1.0 ms, 7.0 ms) (whose sum $2\epsilon_{\text{eff}} = \epsilon_{\text{verifier}} + \epsilon_{\text{gateway}} \sim$ Triangular(2.0 ms, 8.0 ms, 14.0 ms) has mean 8.00 ms and theoretical SD $\sqrt{6} \approx 2.45$ ms, with common-mode regional master reference $\delta_{\text{verifier}} - \delta_{\text{gateway}} =$

0 ms in Federated Egress Gateway Mode), upstream speculative verification durations Δ_{stage} clamped to $[200, 5{,}000]$ ms from LogNormal($\mu = 6.2, \sigma = 0.55$) (theoretical unclamped median 492.7 ms, 5-seed empirical median 491.4 ms, and clamped arithmetic mean 572.7 ± 336.8 ms), and sibling verifier completion offsets following $\Delta t_{\text{sib}} \sim \text{Normal}(\mu = 0 \text{ ms}, \sigma = 22 \text{ ms})$ around T_E—yielding a stochastic clock-/network boundary split across the 148.0 aborts of 119.0 ± 1.1 (80.41%, 5-seed mean ± SD) aborting during Phase 1 concurrent verification and 29.0 ± 1.1 (19.59%) arriving within the $2\epsilon_{\text{eff}}$ cross-node clock-skew window during Phase 2. Because net intra-cluster settlement token transit minus abort margin ($X = \Delta_{\text{real_ipc}} - \Delta_{\text{abort_prop}} \sim \text{LogNormal}(\mu = 0.72, \sigma = 0.18)$ ms, mean 2.09 ± 0.38 ms with Gaussian-copula load correlation $\rho = 0.42$ against $2\epsilon_{\text{eff}}$, analytical variance $6.00 + 0.146 - 2(0.42)(2.449)(0.382) = 5.36$ ms^2 [SD = 2.32 ms], and 0.0% zero-clamping since $\min(2\epsilon_{\text{eff}}) = 2.0$ ms) elapses before $t_{\text{now,earliest}}$ is queried, the residual Phase 2 wait Δ_{wait} pooled over the $2{,}520 - 372 = 2{,}148$ non-aborted committed mutating frames per seed averages 5.91 ± 2.32 ms ($p_{50} = 5.90$ ms, $p_{95} = 9.80$ ms, $p_{99} = 10.84$ ms; when sampling independent non-zero local clock offsets δ_u with $\sigma_{\Delta\delta} = 0.50$ ms across decoupled references, Δ_{wait} is 5.91 ± 2.37 ms with $p_{95} = 9.81$ ms, $p_{99} = 10.96$ ms). Transaction-level causal inversions are counted out of $N = 1{,}000$ transactions when any mutating or dependent frame arrives at the destination ERP ingress boundary ahead of a causally preceding frame ($\mathcal{V}_1 \prec_{\mathcal{V}} \mathcal{V}_2$) or out of physical commit order across concurrent conflicting writes.

B. Comparative Results and Component Ablations

Table I compares Agent-Chronos against four baselines: (1) an **Unmitigated Baseline Swarm**, (2) **Application-Layer Compensating Sagas / Post-Facto Undo** [4], [13], [14] (where compensation RPCs drawn from LogNormal($\mu = 4.90, \sigma = 0.35$) ms [mean 142.53 ± 51.59 ms, conditional $p_{50}/p_{95}/p_{99} = 135.39/233.63/300.82$ ms] are dispatched across all 148.0 aborted transactions per seed, succeeding on 106.0 ± 1.6 [71.62%] reversible writes and failing with an external settlement-lock error after the RPC round-trip on 42.0 ± 1.6 [28.38%] non-reversible writes), (3) a software-only **NTP Hybrid Logical Clock (HLC) Barrier** [8] enforcing dynamic $\Delta_{\text{wait,NTP}} = \max(0, 2\hat{\epsilon}_{\text{NTP}} + \Delta_{\text{abort_prop}} - \Delta t_{\text{ipc}})$ over local NTP daemon error estimates $\hat{\epsilon}_{\text{NTP}} \sim \text{Normal}(\mu = 35 \text{ ms}, \sigma = 8 \text{ ms})$ plus a 5% Bernoulli spike mixture Uniform$(40, 85)$ ms (where half of spikes are reflected in $\hat{\epsilon}_{\text{NTP}}$—raising wait SD to 19.20 ms and preventing abort leaks—and half are uncompensated physical offset spikes on sibling branches that exceed the 25 ms scalar HLC ingress watermark hold window, causing 21.0 ± 1.9 [2.10%] sibling causal inversions and 1.2% of post-settlement dependent reads observing stale state), and (4) a **Chronos Vector-Only Ablation** (removing Phase 2 TrueTime commit-wait pacing while retaining Phase 1 staging, logical vector clocks, and per-destination sequence fencing—serving as an architectural proxy for logical pre-commit staging runtimes such as Atomix [18], Sherlock [17], Speculative Actions [16], and ALAS [19] when deployed across *decentralized cross-node verification topologies*).

In the Unmitigated Baseline, the 195.0 ± 4.2 (19.50%) destination causal inversions decompose into 148.0 ± 0.0 transactions where an unrevoked speculative mutation arrived ahead of its subsequent abort/recovery branch plus 47.0 ± 4.2 pure WAN packet-reordering inversions across valid multi-frame transactions. Ablating TrueTime commit-wait (**Chronos Vector-Only Ablation**) intercepts the 119.0 ± 1.1 (80.41%) Phase 1 aborts before wire release, but leaks the 29.0 ± 1.1 (2.90%) cross-node race aborts arriving within the $2\epsilon_{\text{eff}}$ physical clock-skew window; because each leaked mutation executes ahead of the orchestrator's subsequent recovery write on the same resource key—and because ablating TrueTime physical bounds forces the ingress sidecar to fall back to arrival-order tie-breaking on vector-concurrent colliding sibling writes ($\mathcal{V}_1 \parallel \mathcal{V}_2$, 5.0 ± 0.9 collisions/run under Zipf(1.15)), Vector-Only incurs $29.0 + 5.0 = 34.0 \pm 1.5$ (3.40%) destination inversions. Note that ingress jitter re-sequencing hold latency (8.42 ± 4.90 ms, which includes $+0.03$ ms amortized from sibling-fork holds across the $5.0/2{,}148$ colliding committed frames) matches between Vector-Only and Full Agent-Chronos on the 90 ms WAN link because minimum one-way transit (15 ms) already exceeds $2\epsilon_{\text{eff}} \leq 14$ ms upon arrival ($t_{\text{now,earliest}} \geq t_{F,\text{latest}}$); by contrast, in an auxiliary low-latency intra-region metro interconnect experiment (4 ms base RTT, one-way transit $[1, 3]$ ms $< 2\epsilon_{\text{eff}}$), the ingress physical-time barrier ($t_{\text{now,earliest}} \geq t_{F,\text{latest}}$) actively holds arriving frames for 3.92 ± 1.85 ms to preserve 0.00% inversions.

Full **Agent-Chronos** intercepts 100.00% (148.0/148.0) of aborted mutations pre-wire and eliminates 100.00% (195.0/195.0) of causal inversions, with a mean Phase 2 commit-wait of 5.91 ± 2.32 ms ($p_{50} = 5.90$ ms, $p_{95} = 9.80$ ms, $p_{99} = 10.84$ ms; when remote cross-cloud quorum verifiers require $\Delta_{\text{wan_pre}} = 20$ ms, pre-minting quorum wait adds $+18.5 \pm 4.2$ ms) and 0.00 ms delay on independent read-only queries (94.2% bypass immediately; the 83.0 non-aborted post-settlement dependent reads wait 5.40 ± 1.98 ms via `SAFE_READ_WAIT`). End-to-end, compared to non-speculative sequential execution ($T_{\text{seq}} = t_{\text{exec}} + \Delta_{\text{stage}} = 650.0 + 572.7 = 1{,}222.7 \pm 373.3$ ms mean completion, summing the 650.0 ms execution path and 572.7 ms clamped mean verification duration [median 492.7 ms]), Agent-Chronos overlaps upstream agent synthesis/staging ($t_{\text{prep}} = 560.0$ ms) with verification (Δ_{stage}): across the 85.2% non-aborted committed transactions, completion latency is $T_{\text{spec,commit}} = \max(t_{\text{prep}}, \Delta_{\text{stage}}) + \Delta_{\text{wait}} + t_{\text{rtt}} = 703.6 + 5.91 + 90.0 = 799.5 \pm 278.1$ ms (where $\mathbb{E}[\max(t_{\text{prep}}, \Delta_{\text{stage}})] = 703.6$ ms reflects the $+143.6$ ms upper-tail excess when $\Delta_{\text{stage}} > t_{\text{prep}}$), achieving a net speculative speedup of 423.3 ms (34.6%). Across all 1,000 transactions unconditionally (including the 14.8% aborted branches, which incur $+240.0 \pm 45.0$ ms fallback re-synthesis upon pre-wire revocation, adding $+34.7$ ms unconditional amortized retry overhead), mean completion is 834.2 ± 292.5 ms (an unconditional speedup of 388.5 ms, or 31.8%). When invoking unmodified third-party SaaS APIs without an ingress sidecar via sequential per-resource in-flight pipelining (0% injected HTTP 5xx faults, incurring one 90 ms WAN RTT per dependent same-key write across 46.6% of transactions), mean transaction latency increases by $+42.0 \pm 19.4$ ms (841.5 ms committed / 876.2 ms unconditional) with 0.00% phantom mutations and 0.00% inversions. Evaluating Hardware PTP

TABLE I
EMPIRICAL BENCHMARK AND ABLATION COMPARISON ACROSS $N = 1{,}000$ CROSS-CLOUD MULTI-AGENT TRANSACTIONS PER SEED (5 SEEDS)[a]

Metric (Basis / Granularity)	Unmitigated Baseline	Compensating Sagas [4], [13], [14]	NTP Software HLC [8]	Chronos Vector-Only (Ablation)	Agent-Chronos (Full)
Transactions Evaluated (Per Seed, $n = 5$ Seeds)	1,000	1,000	1,000	1,000	**1,000**
Phantom Tool Mutations (Tx Count, % of 1,000 Tx)[a]	148.0 ± 0.0 (14.80%)	42.0 ± 1.6 (4.20% uncomp.)	0.0 ± 0.0 (0.00%)	29.0 ± 1.1 (2.90% skew leak)	$\mathbf{0.0 \pm 0.0}$ (**0.00%**)
Causal Inversion Rate (Tx Count, % of 1,000 Tx)[a]	195.0 ± 4.2 (19.50%)	195.0 ± 4.2 (19.50%)	21.0 ± 1.9 (2.10%)	34.0 ± 1.5 (3.40%)	$\mathbf{0.0 \pm 0.0}$ (**0.00%**)
Pre-Wire Aborted Revocations (% of 148 Aborted Tx)[a]	0.0 ± 0.0 (0.00%)	0.0 ± 0.0 (0.00%)	148.0 ± 0.0 (100.00%)	119.0 ± 1.1 (80.41%)	$\mathbf{148.0 \pm 0.0}$ (**100.00%**)
Mean Commit-Wait Pacing (Δ_{wait}, Per-Frame SD)[a,b]	0.00 ms	0.00 ms ($+142.53 \pm 51.59$ ms undo)[b]	68.40 ± 19.20 ms	0.00 ms	$\mathbf{5.91 \pm 2.32}$ ms
Commit-Wait Tail Latency ($p_{50}/p_{95}/p_{99}$)[b]	0.00/0.00/0.00 ms	0.00/155.73/220.89 ms (undo)[b]	67.10/98.40/112.60 ms	0.00/0.00/0.00 ms	**5.90/9.80/10.84** ms
Ingress Jitter Re-Sequencing Hold (Per-Frame SD)[a]	0.00 ms	0.00 ms	11.20 ± 6.40 ms	8.42 ± 4.90 ms	$\mathbf{8.42 \pm 4.90}$ ms
Independent Read-Only Bypass (94.2% of 1,680 Reads)	0.00 ms	0.00 ms	0.00 ms	0.00 ms	**0.00** ms
Dependent Read-Only Wait (`SAFE_READ_WAIT`, 5.8% of Reads)[c]	0.00 ms (stale read)	0.00 ms (stale read)	61.20 ± 17.50 ms	0.00 ms (stale read in $2\epsilon_{\text{eff}}$)[c]	$\mathbf{5.40 \pm 1.98}$ ms
Protocol Auditability	None	Application Logs Only	Partial (Scalar HLC)	Logical Vector Only	**100.00% (Dual-Plane)**

[a]Event-count rows report 5-seed mean $\pm$ across-seed SD ($n = 5$ seeds; total abort budget is fixed at 148.0 ± 0.0 Tx/seed); continuous latency rows report 5-seed mean $\pm$ pooled per-frame SD over the 2,148 committed mutating frames/seed.
[b]For Compensating Sagas, $+142.53 \pm 51.59$ ms is the conditional rollback latency (LogNormal($\mu = 4.90, \sigma = 0.35$) ms, theoretical mean 142.77 ms, SD 51.55 ms) across all 148 aborted Tx/seed ($+21.09 \pm 54.37$ ms unconditional across 1,000 Tx), while 0.00/155.73/220.89 ms (undo) reports unconditional rollback percentiles across all 1,000 Tx (85.2% non-aborted at 0.00 ms; conditional $p_{50}/p_{95}/p_{99} = 135.39/233.63/300.82$ ms, where unconditional p_{95}, p_{99} equal conditional $p_{66.2}, p_{93.2}$).
[c]Of the 5.8% (97.4/1,680) post-settlement dependent reads/seed (83.0 non-aborted, 14.4 aborted via HTTP 409), 100% observe stale state under Unmitigated/Saga, 19.6% under Vector-Only ($\Delta_{\text{wait}} = 0$ ms), 1.2% under NTP HLC, and 0.0% under Chronos.

Sidecar Mode ($\epsilon_{\text{PTP},u}$ i.i.d. $\sim$ Uniform(1.0 ms, 10.0 ms), $2\epsilon_{\text{eff}} \sim$ Triangular(2.0 ms, 11.0 ms, 20.0 ms)) similarly achieved 0.00% phantom mutations and 0.00% inversions at a mean commit-wait of 8.84 ± 3.12 ms and ingress hold of 8.76 ± 5.04 ms (reflecting the 15–20 ms upper tail of $2\epsilon_{\text{eff}}$), while user-space L7 staging required only 1.8 MB peak memory across 100 concurrent staged frames and < 0.12 ms CPU overhead per frame. Finally, an auxiliary stress suite confirmed 0.00% phantom mutations and 0.00% inversions under: (i) heavy-tailed Pareto($\alpha = 2.5$) WAN jitter spikes up to 180 ms (where conflicting sibling writes arriving past $\Delta_{\text{jitter_hold}} = 60$ ms are safely rejected with HTTP 409 Stale Sibling Timestamp), (ii) temporally autocorrelated sawtooth GPS holdover (ϵ_u ramping to 12 ms, raising p_{99} commit-wait to 19.40 ms), (iii) 2% injected mid-flight HTTP/2 `RST_STREAM` resets (resolved via tombstone sequence ACKs with zero ingress stalls), and (iv) 16-agent swarm fan-out (121 B unformatted positional / 184 B with optional metadata max raw JSON vector).

V. CONCLUSION AND OPERATIONAL SCOPE

Agent-Chronos establishes a hardware-anchored L7 transport fencing and speculative pacing protocol for cross-cloud multi-agent swarms. By combining two-phase user-space speculative staging, TrueTime commit-wait serialization ($\Delta_{\text{wait}} = \max(0, t_{F,\text{latest}} + \Delta_{\text{abort_prop}} - t_{\text{now,earliest}})$), decoupled swarm-wide causal vectors and per-destination sequence coordinates, and ingress jitter re-sequencing, Agent-Chronos eliminates phantom tool mutations and causal inversions across Chronos-enabled sidecar gateways while delivering a 423.3 ms (34.6%) net speculative latency reduction per committed transaction (388.5 ms, or 31.8%, unconditionally across all transactions) without holding distributed database locks. Functionally, Agent-Chronos guarantees strict client-side egress linearizability and end-to-end causal delivery across sidecar-fronted endpoints rather than internal database strict serializability; when invoking unmodified third-party SaaS APIs that lack an ingress sidecar, the gateway suppresses tombstone sequence ACKs, routes concurrent sibling writes ($\mathcal{V}_1 \parallel \mathcal{V}_2$) through a shared logical egress gateway, and falls back to sequential per-resource in-flight pipelining (+42.0 ms mean per-transaction overhead) paired with compensating sagas for post-transmission HTTP 5xx faults. Operational deployment assumes access to TrueTime, local `/dev/ptp0` hardware clocks, or a federated Chronos egress gateway, alongside bounded active swarm fan-out ($|\text{AgentID}| \leq 16$). Future work will extend Agent-Chronos to dynamic hierarchical vector compression for massive (> 100-agent) swarms and live production trace replay over multi-region enterprise meshes.

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